

Silently Absorbed or Loudly Flagged: Local Falsifiability Governs How Transformers Respond to Planted Errors in Their Own Reasoning

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Abstract

Compounding-error arguments against autoregressive reasoning assume mid-trajectory errors are unrecoverable. We test the premise causally: take multi-hop deduction problems a pretrained transformer solves, inject a single controlled sentence into the model’s *own* proof, remove all external feedback, and formally validate every step of the continuation. Our central finding required catching our own construct error: a perturbation generator that picks facts “off the gold proof’s path” — the natural reading of *wrong* — produces statements that are **entailed-true 92% of the time** in PrOntoQA worlds. Auditing injection truth status inverts the apparent result. True-but-irrelevant interruptions are shrugged off (87% formally valid re-derivation, 0% error adoption). Genuinely false injections split by *local falsifiability*: lies contradicting evidence available within one inference hop are loudly flagged (up to 39% verbalized doubt) and largely survived (64–75% valid re-derivation), while lies that can only be falsified by global entailment checking are **silently absorbed** — 0% verbalized doubt, valid re-derivation collapsing to 23–34%, and the planted fact entering the model’s subsequent derivation in 29–60% of runs. All falsehoods cause damage exceeding a benign-paraphrase control ($p < .03$ at every injection position); a formal step-validator separates genuine re-derivation from goal parroting, which inflates naive final-answer “recovery” by 10–20 points. The gradient is parametric: placing the falsifying evidence exactly k inference hops away yields monotonically falling doubt ($.39 \rightarrow .32 \rightarrow .23 \rightarrow .00$) and recovery as k grows. Three further results bound the claim. Recovery survives a question-format condition in which the queried statement’s truth value is unknown (57% formally validated, so it is not goal parroting). The regime sets the price: on GSM8K and chained arithmetic, where intermediates are load-bearing, absorption reaches 88–100% — and the GSM8K case shows models re-check what they can re-read in context, not what they must recompute. Scale attenuates absorption (32B recovers from hard-to-check lies at $\sim 3\times$ the 7B rate) but not silence, and a second model family (OLMo-2) replicates the recovery gradient while verbalizing nothing — flagging is training-dependent, recovery is not. All rates are measured on a doubly filtered cohort (instances the model solves, whose unperturbed proofs formally validate) and are upper bounds. For the compounding debate the upshot is sharpened rather than settled: per-step error *absorption* is real precisely when errors are hardest to check — the dangerous case — while cheaply-checkable errors are not absorbed. Off-path $\neq$ false is a construct trap for any perturbation study on entailment-dense tasks; we release the truth-status audit with the pipeline.

1 Introduction

A stylized argument pattern, assembled from ingredients in the compounding-error literature [1, 3], holds that per-token commitment makes long autoregressive generations collapse: if each step errs with probability ε and errors are independent, constant-rate, and unrecoverable, success decays like $(1 - \varepsilon)^T$. The *unrecoverable* premise is the load-bearing one, and it is usually attacked externally, via verifiers and tools. We ask the internal question: when a single error is planted in a pretrained transformer’s own reasoning trajectory, with all external feedback removed, what does the model’s subsequent computation actually do with it?

Prior interventional work perturbs visible chain-of-thought and reads out behavioral robustness [12]; redundancy and self-repair results [2, 4] suggest internal robustness mechanisms exist. What is missing is (i) *semantic control over what the perturbation actually is* — wrong relative to what? — and (ii) a behavioral

metric strong enough to distinguish re-derivation from reaching a goal stated in the prompt. Both gaps turn out to be load-bearing. Controlling them yields a result neither the fragility nor the robustness literature predicts cleanly: the model’s response to a planted error is governed by how cheaply the error can be falsified against its local context.

Contributions. (1) A perturb-and-trace protocol on the model’s own solved rollouts with a *formal step-validator*: each continuation is classified as valid re-derivation, poisoned (a later step derivable only through the planted error), parroted (goal reached via an underivable step), or derailed. Naive final-answer accuracy overstates recovery by 10–20 points. (2) A *truth-status audit* exposing a construct trap: in entailment-dense worlds, “off the gold path” injections are usually entailed-true (92% here). Conclusions about error recovery invert once injections are audited for actual falsity. (3) Six perturbation families spanning a falsifiability gradient — benign paraphrase, true interruption, irrelevant rule, contradiction of an established fact, negation of the imminent step, and unfalsifiable-without-global-checking false fact — each $n=130$ runs per injection position (35 for the last, availability-limited). (4) The finding: *silent absorption of hard-to-check lies* (0% doubt, 29–60% poisoning) vs. *loud flagging of cheap-to-check lies* (18–39% doubt, 0% measurable poisoning, 64–75% valid re-derivation), with true interruptions shrugged off and all falsehoods significantly worse than the benign control.

2 Setup

Task and model. PrOntoQA-OOD [5] generates multi-hop deductions over fictional ontologies in machine-parseable sentences. We use 500 instances (4–6 hops), few-shot prompting in the canonical proof style, and frozen Qwen2.5-7B-Instruct (greedy decoding). The model solves 91% of instances; the analysis cohort is the solved subset whose gold rollout itself passes the formal validator (84%; $n=168$ instances, $n=130$ runs per family×position cell).

Protocol. We parse the model’s own rollout into sentences, select an intermediate derived fact at an early/mid/late position, replace it, and let the model continue greedily with no feedback, no critique prompt, and a fixed budget. Hidden states are cached at five depths for all rollouts. For each instance and position, a null ensemble of $n=8$ temperature-0.8 continuations from the same prefix with the *correct* step (filtered to those remaining correct) calibrates all latent comparisons.

Perturbation families and the falsifiability gradient. (a) **Benign paraphrase**: the correct step with a discourse marker prepended — semantically identical, controls for any-disruption effects. (b) **True interruption**: an entity-category fact off the gold proof’s path but *entailed by the world* (see §4 — this family was designed as “wrong” and discovered to be true). (c) **Distractor**: a fabricated rule irrelevant to the entity. (d) **Contradiction**: negation of the most recently *established* fact — false, and in direct conflict with stated context (checkable at distance zero). (e) **Negated step**: negation of the fact the model was about to derive — false, not conflicting with any stated sentence, but falsifiable by a one-hop inference from context. (f) **Unfalsifiable-locally false fact**: an entity-category assertion verified non-entailed under the world’s rule closure — detectable only by global checking. Constructions (d)–(f) are verified false (and (b)’s truth verified) against the rule closure, per-injection.

3 Metrics

Formal validation. A continuation is a **valid re-derivation** iff every entity-fact sentence is entailed by the true world state (premises plus prior valid derivations) under the transitive closure of the question’s rules, and the final statement matches the target. **Poisoned**: some step is derivable *only* through the planted atom. **Parroted**: target reached despite an underivable intermediate step. **Derailed**: wrong final statement. Discourse markers are stripped before parsing (a bug our benign control caught: the model adopts injected style, which broke naive parsing); unparsed runs are excluded and counted. Two scope notes, stated

plainly: closure entailment permits step-skipping, so we separately measured derivation-free goal-jumps (4.7–9.2% of valid re-derivations at early/mid positions; at late positions gold itself has no remaining intermediate steps); and poisoning is mechanically measurable only for *positive category* injections — the grammar derives nothing from negated facts — so 0% poisoning for families (d)–(e) is a measurement bound, not evidence of resistance.

Doubt. A lexical detector (wait/however/but/actually/contradict/...) over the continuation. Crude, but the cross-family contrasts it yields exceed 5σ .

4 The truth-status audit

Our original “wrong-category” generator guaranteed injections were off the gold proof’s path. Auditing every injection against the world’s rule closure showed **92% were entailed-true**: PrOntoQA entities carry second premises and distractor rule branches, so most off-path categories are still true of the entity. Splitting that family’s outcomes by audited truth status inverts the headline (Table 1): true injections are shrugged off; the genuinely false minority is absorbed. Every conclusion below uses truth-audited families. We flag this as a general trap: in any entailment-dense benchmark, “not in the gold derivation” is a property of the *proof*, not of the *world*.

Injection (audited)	n	Valid re-deriv.	Poisoned	Parroted
entailed-true	355	.868	.000	.082
genuinely false	35	.257	.457	.200

Table 1: The same generator’s injections, split by audited truth status (wrong-category family, all positions pooled). The apparent “recovery from lies” was recovery from *true statements*.

5 Results

5.1 The falsifiability gradient

Table 2 is the main result. Reading top to bottom: benign disruption costs 8–23 points of valid re-derivation relative to the construction ceiling (unperturbed gold = 100% by cohort definition) — *any* mid-proof interruption has a price. True interruptions and contradictions of stated facts cost no more than benign ($|z| < 1.5$, n.s.). Genuine falsehoods cost significantly more than benign at every position (negated step: $z = -2.18 / -3.20 / -3.64$, $p = .029 / .0014 / .0003$) — and split sharply by checkability:

Silent absorption. For falsehoods detectable only by global entailment checking, the model verbalizes doubt in 0/70 early+mid runs, completes a confident-looking proof, and incorporates the planted fact into its derivation in 29–60% of runs. **Loud flagging.** For falsehoods refutable by a one-hop check against context, verbalized doubt jumps to 18–39% (the highest of any family; contradiction-vs-true-interruption doubt contrast $z = 5.5$, $p < 10^{-7}$) and valid re-derivation stays at 64–75%. The model’s error response is gated not by falsity but by the *cost of the check*. The negated-step-vs-benign deficits survive Holm correction across the three injection positions (adjusted $p = .029 / .0028 / .0009$); per our pre-registration, the original wrong/early cell is the designated primary contrast and all other behavioral contrasts are exploratory.

The gradient is parametric: a dose-response in inferential distance. Negating the fact the model will derive k hops downstream places the falsifying evidence exactly k inferences away. At early injection ($n = 130/\text{cell}$ for $k \leq 3$), verbalized doubt falls — .392 ($k=1$) $\rightarrow$.323 ($k=2$) $\rightarrow$.231 ($k=3$) — then *plateaus* at $k=4, 5$ (.229, .246; $n=96, 69$ — only longer chains admit distant targets, a selection caveat), with the global-only construction far below the plateau (.000). Validated recovery follows the same decline-then-plateau shape (.646 $\rightarrow$.562 $\rightarrow$.531 $\rightarrow$.594 $\rightarrow$.594). The pre-registered monotone prediction is confirmed over $k=1-3$ and amended to decline-then-plateau beyond. At 32B the curve shifts up and compresses: recovery

Family	Inj.	Valid re-deriv. [95% CI]	Poisoned	Parroted	Doubt
benign paraphrase	early	0.77 [0.69,0.83]	—	0.05	0.08
benign paraphrase	mid	0.82 [0.74,0.87]	—	0.05	0.16
benign paraphrase	late	0.92 [0.85,0.95]	—	0.02	0.05
true interruption	early	0.82 [0.75,0.88]	0.02	0.07	0.01
true interruption	mid	0.75 [0.67,0.82]	0.07	0.12	0.03
true interruption	late	0.86 [0.79,0.91]	0.03	0.09	0.02
distractor rule	early	0.73 [0.65,0.80]	0.00	0.10	0.02
distractor rule	mid	0.72 [0.63,0.79]	0.00	0.14	0.03
distractor rule	late	0.76 [0.68,0.83]	0.00	0.15	0.03
contradiction (dist. 0)	early	0.73 [0.65,0.80]	0.00	0.11	0.23
contradiction (dist. 0)	mid	0.79 [0.71,0.85]	0.00	0.02	0.27
contradiction (dist. 0)	late	0.85 [0.78,0.90]	0.01	0.02	0.18
false, 1-hop checkable	early	0.65 [0.56,0.72]	0.00 [†]	0.09	0.39
false, 1-hop checkable	mid	0.64 [0.55,0.72]	0.00 [†]	0.05	0.25
false, 1-hop checkable	late	0.75 [0.67,0.81]	0.00 [†]	0.05	0.18
false, globally checkable only	early	0.34 [0.21,0.51]	0.29	0.20	0.00
false, globally checkable only	mid	0.23 [0.12,0.39]	0.60	0.17	0.00
false, globally checkable only	late	0.29 [0.16,0.45]	0.43	0.23	0.06

Table 2: Formally validated outcomes, Qwen2.5-7B, $n=130/\text{cell}$ (35 for the last family, availability-limited to $\sim 30\%$ of instances; a within-subset control shows the negated-step family performs identically on those instances, .676 vs .677, so the gap is a construction effect, not instance selection). [†]Poisoning is not measurable for negated injections (see Metrics). “—”: not applicable (injection true by construction).

.699–.753 at every k , doubt .237/.151/.161 — scale lifts the curve rather than steepening it. Checkability is a quantity, not a category; its behavioral effect saturates a few hops out.

Prompting does not fix the dangerous case. A verification instruction (“before using any stated fact, verify it follows from the rules; if a statement conflicts, point this out...”) was pre-registered to cut silent absorption by $\geq 50\%$. It does not: poisoning on globally-checkable falsehoods falls only $\sim 17\%$ relative (pooled .438 $\rightarrow$.362, $n=105/\text{arm}$), and doubt on those lies stays near zero. What the instruction *does* do is nearly double verbalized doubt on already-checkable lies (.392 $\rightarrow$.585 early) while *reducing* their validated recovery (.646 $\rightarrow$.554; mid .638 $\rightarrow$.469) — the model spends its budget second-guessing what it could already check, echoing the reasoning-trained model’s reactivity cost. Verbalization is cheap to induce; checking what is locally unfalsifiable is not.

The doubt instrument is not a regex artifact. A Qwen2.5-32B judge, prompted to classify each mid-injection continuation for expressed doubt or self-correction, agrees with the lexical detector on 76–100% of runs per family and *sharpens* the contrast: judge-rated doubt is .285 (1-hop falsehood) and .246 (contradiction) vs. .015 (true interruption) and .000 (globally-checkable falsehood); the lexical detector’s largest deviation (benign paraphrase, .162 lexical vs. .092 judge) reflects regex false positives on discourse markers, in the direction that *understates* our contrast.

5.2 The regime supplies recoverability: three tasks, one protocol

The same mid-trajectory corruption protocol across task structures (7B):

Task	Recovery	Absorption	n
PrOntoQA (redundant derivations)	.53–.86	.00–.60*	130/cell
GSM8K (natural-language math)	.088	.877	57
Chained arithmetic (no redundancy)	.00–.02	.99–1.00	99/cell

(*by family; GSM8K cohort is format-filtered — correct greedy solutions with ≥ 3 explicit computations, 57/250 — selecting multi-step explicit-arithmetic problems; corruption at the middle computation.) Redundant tasks afford re-derivation; tasks where each intermediate is load-bearing convert nearly every absorbed error into a wrong answer. Notably, the corrupted GSM8K value *is* locally checkable — one recomputation — yet absorption dominates: the checking that drives the PrOntoQA gradient operates over *stated context facts*, not over recomputation. Models re-check what they can re-read, not what they must re-derive numerically. “Recoverability” is a property of the task’s derivation structure and the check’s modality, at least as much as of the model.

5.3 Recovery is not goal-anchor re-reading

Two ablations target the concern that recovery is just attention re-reading the stated target. **(a) Question format:** rephrasing instances as “True or false: s ?” — with s the entailed target on half the instances and its negation on the other half — removes the parrotable goal (the statement is visible; its truth value is not). Validated recovery (correct verdict + formally sound post-injection chain) remains 57% under 1-hop falsehoods and 64% under benign paraphrase ($n=35/36$; QA gold solve rate 55%, so the cohort is smaller and harder). Attenuated, not collapsed: goal-anchoring is not a sufficient explanation. **(b) Attention knockout** on the goal-statement span collapses *both* the falsehood arm (.015) *and the benign arm* (.046) — the masking breaks the task (without a visible target, proof termination is undefined), so it cannot isolate the recovery mechanism; we report it as a negative design result. Its salvageable secondary: goal-masked continuations remain chain-sound at identical rates for benign and falsehood arms (.45 vs .42), i.e., goal access matters for derivation quality generally, not differentially for error recovery.

5.4 Latent trace

A null-calibrated functional divergence (ρ_h^{func} : Jensen–Shannon divergence between perturbed and gold next-token distributions at sentence boundaries, normalized by the correct-null ensemble’s spread; LM-head computation verified exactly against live logits; instance-specificity verified by a shuffled-gold control, $1.49\times$) shows a consistent shape at well-powered horizons (early/mid injections, $h \leq 4$, $n \geq 106$ per point): a large spike at the first post-injection boundary, decaying toward but remaining above the null band ($\rho_4^{\text{func}} \approx 1.2\text{--}2.3$ across families). We report this as descriptive only, for two reasons measured in our own data. First, the null ensemble is temperature-sampled while rollouts are greedy, a decoding mismatch that inflates the denominator by an unquantified factor. Second, and decisively: the *benign paraphrase* family shows the largest early-injection divergence of any family ($\rho_4^{\text{func}}=2.32$ vs. 1.9–2.2 for genuine falsehoods) — the model adopts the injected discourse style for the rest of its proof, and the functional metric tracks that surface change as strongly as it tracks semantic damage. As instrumented, ρ^{func} cannot distinguish a lie from a stylistic tic. Three follow-up controls close the question. *Matched decoding*: sampling perturbed continuations at the null’s temperature ($T=0.8$, $n=4/\text{site}$) removes the greedy-vs-sampled confound, and benign $\approx$ falsehood persists (h1: 22.4 vs 16.3; h4: 1.36 vs 1.58) — the null result is not a decoding artifact. *Ensemble stability*: across reasonable null configurations ($T \in \{0.6, 0.8, 1.0\}$, $n \in \{4, 8, 16, 32\}$) the instance-level ρ^{func} median ranges 1.9–7.9 with interquartile ranges spanning two orders of magnitude — the ratio is unstable at feasible ensemble sizes. Late-injection horizons beyond $h=2$ collapse to $n \leq 12$ and support no conclusions. Raw hidden-state geometry tells the same story: null-calibrated cosine divergence at the final layer is statistically indistinguishable between benign paraphrase (median 2.57) and verified falsehoods (2.52–2.99) at well-powered horizons — surface form dominates content in geometric trajectory comparisons too. (A layer-0 lexical control was attempted but is degenerate at sentence boundaries, where the position is always the period token; a pooled-sentence control is future work.) We accordingly make *no* latent-recovery claims: the paper’s evidence is behavioral and formally validated.

5.5 Scale attenuates absorption but not silence; the gradient generalizes across model families

Running the truth-audited falsehood families across Qwen2.5 sizes answers the scale question — and partially falsifies our pre-registered prediction (we predicted absorption persists; it persists *attenuated*):

Model	globally-checkable falsehood			1-hop falsehood
	valid	poisoned	doubt	doubt
Qwen2.5-1.5B	.15-.46	.31-.62	.000	.000-.018
Qwen2.5-7B	.23-.34	.29-.60	.000-.057	.18-.39
Qwen2.5-32B	.60-.73	.10-.37	.033-.067	.13-.28

Recovery from hard-to-check lies roughly triples from 7B to 32B and poisoning roughly halves ($n=30\text{--}35/\text{cell}$ for this family) — scale buys checking. But the *silence* is scale-invariant: at every size, absorbed lies go un verbalized. Separately, the pre-audit “wrong-category” sweep (predominantly true interruptions) shows robustness to true interruptions also scaling (.60-.67 $\rightarrow$.75-.86 $\rightarrow$.88-.96; parroting .11-.20 $\rightarrow$.07-.12 $\rightarrow$.01-.02).

A second model family, OLMo-2-7B-Instruct ($n=76/\text{cell}$; solve rate .69), replicates the *recovery* gradient (benign .72 $\geq$ 1-hop falsehood .53 at mid; poisoning concentrated in globally-checkable falsehoods, .22-.33 at $n=9$) but **not the flagging**: OLMo verbalizes essentially nothing in any family (.00-.026), recovering from checkable lies as silently as Qwen-1.5B does. The behavioral checkability gradient appears model-general; *verbalization of detected errors is model- and training-dependent* — recovery does not require verbalization even within a single model.

The training axis is isolated by DeepSeek-R1-Distill-Qwen-7B (same backbone, RL reasoning training; injection into the visible proof body, think block stripped for validation; 29% solve, 8–16% unparsed, reported per pre-registration). Pre-registered prediction — doubt rises, recovery flat — confirmed: visible flagging of 1-hop falsehoods reaches **.672** ($n=122$; vs. .25 Qwen-7B, .00 OLMo) while validated recovery stays flat (.689 vs. .638), and the flagging is content-selective (benign paraphrase: .008). The verbalization spectrum — silent (OLMo, Qwen-1.5B) $\rightarrow$ moderate (Qwen-7B/32B) $\rightarrow$ loud (R1) — at roughly constant recovery supports a clean separation: *flagging is trained; recovery is intrinsic*. (R1’s private channel is uninformative here: 100% of its gold think blocks contain doubt vocabulary at baseline — a ceiling. R1 also handles the benign style perturbation *worse* than its base model, .549 vs. .815: reasoning training buys error verbalization at some cost in disruption robustness.)

6 Limitations

Task coverage now spans two synthetic families plus GSM8K (format-filtered cohort, 57/250 — selection toward multi-step explicit arithmetic; Lean theorem proving is the next arm); two base-model families (Qwen2.5, OLMo-2; a reasoning-tuned arm is in progress), greedy gold rollouts. The Prove-format prompt states the target; the validator classifies parroting separately and the QA-format condition shows recovery survives without a parrotable goal, but goal-directedness still aids derivation quality generically (knockout secondary result). The globally-checkable-only falsehood family has $n=35/\text{cell}$ and exists only on $\sim 30\%$ of instances (a within-subset control rules out instance selection); the QA condition has $n=35\text{--}36/\text{cell}$. “Local falsifiability” is operationalized at two levels (distance 0–1 vs. global); the parametric inferential-distance version is future work. Latent and geometric metrics are reported as negative/descriptive results (§5.2). Closure-based validity permits step-skipping (quantified at 4.7–9.2%). One seed of null sampling; behavioral results use deterministic decoding throughout.

7 Related work

CoT interventions and error recovery. von Recum et al. [12] perturb visible CoT in reasoning-tuned models and find behavioral robustness with early-intervention degradation and doubt/length effects; our truth-status audit suggests such interventions need per-injection falsity verification for “adversarial” conditions to mean what they claim. Yee et al. [6] precede our re-derivation/parroting split with a faithful/unfaithful recovery dichotomy via manual annotation; our delta is automated formal validation, semantic (vs. numerical) error control, the truth-status audit, and the falsifiability gradient. Their finding that recovery is more frequent for *obvious* errors independently anticipates checkability-gating. Lanham et al. [7] introduced adding-mistakes interventions for faithfulness; Turpin et al. [8] showed stated reasoning can omit operative causes — consistent with silent absorption. Zhang et al. [9] showed models commit to early

mistakes they can separately recognize; our results locate when that recognition is and is not deployed in-rollout. Huang et al. [10] found intrinsic self-correction weak at the multi-turn level; silent absorption is its single-pass analogue, while flagging shows the capacity exists when the check is local. **Self-repair and redundancy.** Lad et al. [2], McGrath et al. [4] document structural self-repair; silent absorption is the complementary failure: robustness machinery that smooths over content errors it cannot cheaply check. **Trajectory geometry.** Sun et al. [11] characterize reasoning as step-specific representational trajectories with correctness signals; our style-propagation result (benign $\approx$ falsehood in both functional and geometric divergence) is a caution for cross-trace comparisons in that line. **Compounding.** Dziri et al. [1], LeCun [3] supply the ingredients of the compounding frame our results sharpen: per-step absorption is real exactly when errors are locally unfalsifiable, and task derivation structure (§5.3) determines the price.

Reproducibility

Single 8×H200 node, ~8 GPU-hours total. Pre-registered plan (PLAN.md, written before the first run), all scripts, per-run JSONL logs, truth-status audit, and result JSONs accompany the paper (experiments/09_latent_recovery/

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